THE KUIPER BELT AND THE OORT CLOUD

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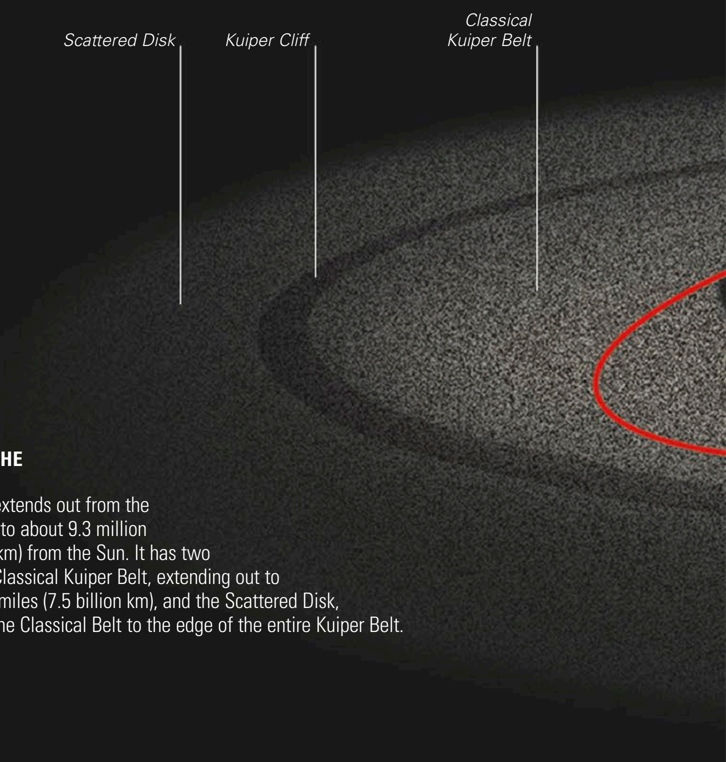
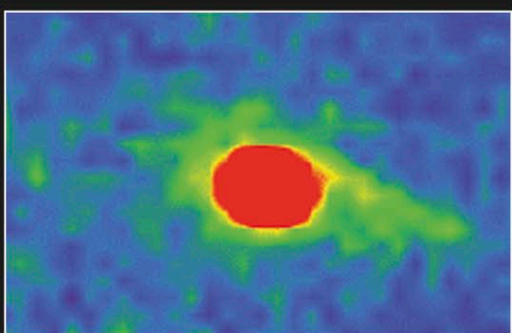
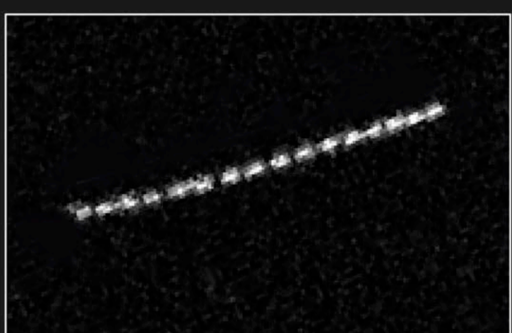
BEYOND THE ORBITS of the giant planets, the solar system is surrounded by billions of small worlds made mostly of various kinds of ice. The inner ones orbit within the Classical Kuiper Belt and are known as Kuiper Belt Objects (KBOs), while others beyond that occur in the Scattered Disk. Some of these icy bodies are the size of small planets, and one—Pluto—was originally classified as a planet in its own right. Beyond the Kuiper Belt lies an enormous halo of smaller icy bodies known as the Oort Cloud. Believed to contain trillions of objects, the Oort Cloud is the source of many of the comets that visit the inner solar system.



LOCATION OF THE KUIPER BELT
The Kuiper Belt extends out from the orbit of Neptune to about 9.3 million miles (15 billion km) from the Sun. It has two subregions: the Classical Kuiper Belt, extending out to about 4.7 billion miles (7.5 billion km), and the Scattered Disk, stretching from the Classical Belt to the edge of the entire Kuiper Belt.



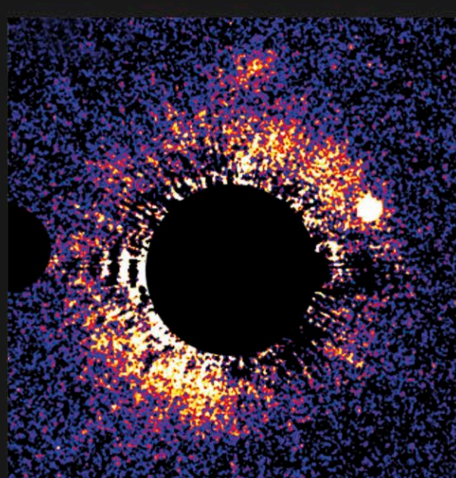
CHIRON
Discovered in 1977, Chiron is the prototype of a group of icy bodies following orbits around those of Saturn and Uranus. Known as Centaurs, they are thought to be Scattered Disk Objects that have been pulled inward by interactions with Neptune's gravity. They may go on to become short-period comets.



QUAOAR
The multiple exposures in this image show Quaoar moving across the sky. Discovered in 2002, it has an estimated diameter of around 730 miles (1,170 km), about half that of Pluto, but Quaoar is much denser than Pluto, indicating that it contains more rock than ice.

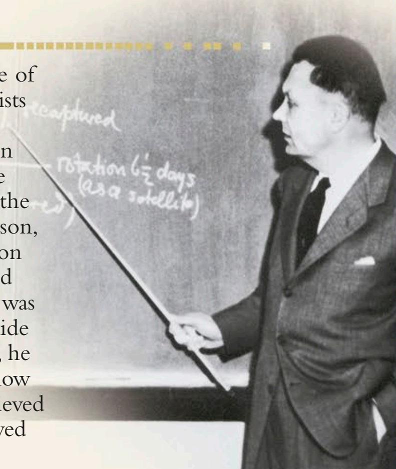
DWARF PLANETS

On its discovery in 1930, Pluto was designated as the ninth planet from the Sun, though at the time there was no formal definition of the term "planet." The subsequent discovery of several Pluto-sized objects in the Kuiper Belt made it clear that Pluto was merely the first Kuiper Belt Object to be discovered. The realization that the Kuiper Belt is populated by numerous bodies, ranging from Pluto-sized objects down to smaller sizes, meant that any definition of a planet could not be based on size alone because there is no natural break in the size distribution. In 2006, the International Astronomical Union (IAU) made a formal definition of the term "planet," stipulating three essential criteria: it must orbit the Sun, it must be massive enough for its own gravity to have pulled it into a round shape, and it must have cleared the neighborhood around its orbit. Pluto fails the third criterion because it shares its orbital space with several objects of comparable size and also crosses the orbit of the vastly more massive Neptune. Pluto was therefore no longer officially a planet. Although this decision was controversial, it is clear that if Pluto had been discovered in 2010 instead of 1930, it would never have been called a planet in the first place. However, the IAU coined a new term, dwarf planet, to denote a body that fails only the third criterion for planet definition. Thus Pluto, Eris, Haumea, Makemake, and probably a few other large KBOs are dwarf planets, as is the largest asteroid, Ceres (see p.175).



EXTRASOLAR DEBRIS DISK
Several Kuiper Belt-like structures have been found around other stars that are thought to be debris left over from the processes of planet formation. The disk around the billion-year-old HD 53143 (shown right), a cool star about 60 light-years from Earth, stretches to around 10.2 billion miles (16.5 billion km) from its central star—roughly the same diameter as our Kuiper Belt and Scattered Disk.

GERARD KUIPER



Gerard Kuiper (1905–1973) was one of the most influential planetary scientists of the 20th century. After studying astronomy at the University of Leiden in the Netherlands, he moved to the United States in 1933. He founded the Lunar and Planetary Institute at Tucson, Arizona, in 1960, and later worked on early planetary probes. He discovered the moons Miranda and Nereid and was also the first to identify carbon dioxide in the atmosphere of Mars. In 1951, he proposed the existence of what we now call the Kuiper Belt, although he believed that its existence had been a short-lived phase of the early solar system.